

Personal CV



Base Information

Name: Zeng Junyan TEL: (+86) 18986010068
E-mail: jzengag@connect.ust.hk Date of Birth: October 1999

Education

Graduation: 07/2022

The Hong Kong University of Science and Technology

09/2018 - 07/2022

Bachelor of Science in Computer Science

- HKUST Entrance Scholarship 09/2018

Key Project Experience

MoE48B/565B Model Training Efficiency Optimization

02/2025 - Present

- Assisted CUDA kernel developers in implementing the Swiglu FC2 Fuse solution.
- Utilized custom Torch functions, Expert Parallel, and Cuda Kernel Fuse technologies to optimize the training efficiency of the MoE48B/565B model to reach an MFU of 30%, reducing training time and costs.

Multi-Modal Large-Model Training and Deployment

10/2024 - Present

- Based on the pre-trained MoE6B/85B LLM, supported the voice embedding, encoder, and adaptor structures for the multi-modal large-scale model.
- Achieved large-scale training on an 1152-card cluster using technologies such as Megatron Tensor Parallel, F1B2 Pipeline Parallel.
- Successfully launched the model for internal use, promoting its application in various business scenarios.

Development and Optimization of Meituan's Text-to-Speech (TTS) Large-scale Model

04/2024 - 10/2024

- Supported the expansion of the TTS model structure from 500M parameters to 1.5B parameters.
- Utilized the Varlen Attention Llama structure and loss-masked DiT structure for efficient parameter expansion and large-scale model training.
- Processed the filtering and packaging operations of 5 million hours of 1.5PB training data, significantly improving the performance of the TTS model.

Decision-Making Service Data Acquisition

12/2022 - 03/2023

- Developed SQL-based heterogeneous data joint-query component using Calcite, integrating hive-spark, s3, and non-table data sources.

Skills and Certificates

- Programming Languages: Proficient in C++, Java, Python, Golang.
- Data Tools: Skilled in Spark, Doris, Tair, ES, Flink, Kafka for big data processing.
- Model Training: Expert in Megatron framework, Torch optimization, and performance analysis.
- Other Skills: Familiar with voice data pre-processing, LLM structures, and data analysis charts.
- Language Ability: Good English communication and reading skills.

Practice Experience

Meituan (Beijing)

08/2022 - Present

Algorithm Engineering Back-end Development

- Led algorithm engineering projects in supply chain optimization and large-model training.
- Implemented system monitoring, caching, and circuit-breaking mechanisms for operational efficiency.

Tencent (Shenzhen)

06/2021 - 08/2021

Management System Back-end Development (Intern)

- Ensured content compliance for network cloud platform audit system.

Areas of Interest

- Model Pruning and Distillation: Focus on reducing model running costs through pruning and distillation to expand their real-world application areas.
- Modal Capabilities of Multi-Modal Models: Aim to enhance multi-modal models' understanding of voice and spatial information for better personal assistant.
- Model Security and Interpretability: Seek to strengthen model security and interpretability to enable applications in high-reliability fields.